

Highlights

Two bases for a voxel, the medium and the wavefield: the error of elastic volume-integral scattering, isolated on a space-filling layer of cubes

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- A voxel needs two bases: one for the medium and one for the wavefield
- On a layer of cubes the discretisation error is isolated, with no shape error
- The order is $\min(2p+2, 2r+2)$ for field degree p and medium degree r
- The single-site T-matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

Two bases for a voxel, the medium and the wavefield: the error of elastic volume-integral scattering, isolated on a space-filling layer of cubes

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Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral (coupled-dipole) methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves, which cannot be measured where no exact solution exists. We answer on a layer of voxels: because cubes tile space, n planes of voxels are the layer itself, with no shape error, and its exact solution isolates the discretisation error. The source-averaged coupling equals the continuum's exactly; the cube's single-site T -matrix follows from distributional moments of the Green's tensor. With the wavefield expanded to Legendre degree p and the medium's contrast to degree r in each voxel, the scheme converges at order $\min(2p + 2, 2r + 2)$, with leading error $c_p(kd)^{2p+2}$ in closed form: $c_0 = 1/12$, $c_1 = 1/720$, $c_2 = 1/100800$; the orders hold for SV and SH waves and at oblique incidence, to $p = 2$. The single-site T -matrix's second-order error equals the relative projection error of the medium at degree $\min(p, r)$, exactly in the long-wave limit: the two bases must be matched,

and the error is known from the medium before anything is solved. Where the medium is piecewise constant on the voxels it vanishes: in a randomly stratified layer one first-moment voxel per layer reaches 10^{-6} . A smoothly graded sphere converges at second order against its exact solution. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background, which the model prescribes. The other is the *wavefield*: the displacement and strain

17 inside the voxel, which the calculation must find. Each is expanded in a basis
 18 on the cell, and the two bases are separate choices. Here both are Legendre
 19 polynomials, the wavefield to degree p and the contrast to degree r ; the
 20 usual scheme, a uniform field in a cell of constant properties, is $p = r = 0$.
 21 This paper asks which basis each of the two needs, and what error a given
 22 pair (p, r) leaves. The answer decides the cost of everything built on it,
 23 because a voxel that represents its contents more accurately can be larger,
 24 and the number of voxels, not the size of each one's state, sets the cost of
 25 the multiple-scattering solve.

26 *Why the question is hard to answer.* For the media such a scheme is built for,
 27 there is no exact solution. Its error cannot be measured against the answer.
 28 Comparison with another numerical method mixes the errors of both. And a
 29 refinement study can mislead, because a voxel scheme can converge smoothly
 30 to the wrong limit. §4 shows how that happens with the most natural choice
 31 of coupling, the Green's tensor evaluated between voxel centres: it carries
 32 a local term whose size per unit volume does not shrink as the voxels do,
 33 and unless the single voxel's response is matched to cancel it, refinement
 34 converges to the wrong answer without revealing it.

35 *The method: a layer of voxels.* This paper therefore answers the question
 36 in the simplest configuration in which a voxel scheme runs in full (every
 37 voxel, every coupling, the whole multiple-scattering series, at normal and
 38 at oblique incidence) and the exact answer is nevertheless known: a layer
 39 filled with voxels. Because cubes tile space, a layer of identical voxels does
 40 not approximate a homogeneous layer; it *is* one. Any difference between
 41 its reflection and transmission and those of the continuous layer is therefore

42 error of the discretisation alone, with no modelling error and no shape error
 43 behind which to hide. A sphere, the usual test object for volume-integral and
 44 discrete-dipole methods [1], cannot give this: its error mixes the discretisation
 45 with the staircase approximation of a curved surface by cubes. The layer
 46 plays the part of the patch test in finite elements: reproducing a uniform
 47 medium correctly is necessary, though not sufficient, for a scheme to be
 48 trusted in a heterogeneous one. The experiment is then made harder in
 49 three steps, each with an exact solution of its own: a randomly stratified
 50 layer, in which every cell is homogeneous but the cells differ; a layer whose
 51 properties vary smoothly within every cell, which is where the basis for the
 52 medium is tested; and a sphere whose contrast falls smoothly to zero at its
 53 surface, which carries the layer's prediction into three dimensions without a
 54 staircase. What the layer establishes about the scheme's two ingredients, the
 55 coupling between voxels and the response of a single voxel, are properties of
 56 the ingredients, which are the same in any medium. A laterally heterogeneous
 57 medium remains to be tested.

58 *The answer.* (1) *The order is set by the weaker basis.* The scheme converges
 59 at order $\min(2p + 2, 2r + 2)$ in the cell size (§9). A contrast held constant in
 60 each cell therefore caps every voxel at second order in a medium that varies
 61 within the cells, however rich the basis of its field; and a contrast of higher
 62 degree than the field is wasted. (2) *The leading error is in closed form, for*
 63 *every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$
 64 with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§8); the orders 2, 4 and 6 of
 65 $p = 0, 1$ and 2 hold for P, SV and SH waves at normal and oblique incidence.
 66 For the collocation scheme, the uniform-field voxel in its simplest form, the

67 error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission (§7), and the
 68 opposite signs explain why no correction of the single site can improve both
 69 (§11). (3) *The error that remains is the single site's, and it is the projection*
 70 *error of the medium.* The single-site T -matrix sums the rescattering within a
 71 voxel to all orders, and the coupling between voxels is exact, so the multiple
 72 scattering itself is not approximated. But the T -matrix is that of the voxel as
 73 represented: its contrast projected to degree r and its field held to degree p .
 74 When the scattered field is separated by its order in the contrast, the error
 75 of the second-order term, in which the field scattered once within a voxel
 76 is scattered there again, equals the *relative projection error* of the medium:
 77 the fraction of the squared norm of its profile that polynomials of degree
 78 $\min(p, r)$ on the cells cannot represent (Eq. (38)). The equality is exact in
 79 the long-wave limit, where it is derived, and holds to within 1.3% for all nine
 80 pairs with $p, r \leq 2$ in the experiments (§9). Splitting that term confirms
 81 whose error it is: the part in which both scatterings lie in one voxel carries
 82 the error, and the part between voxels at most 0.2% of it. It shows what
 83 the wavefield basis is for: it must carry what the medium imprints on the
 84 field, not only the wave, which at these frequencies varies little across a cell.
 85 (4) *The two bases must be matched, and the error is known in advance.* The
 86 relative projection error is a property of the medium and the grid alone, so
 87 the error of a calculation is known before anything is solved, and the cell size
 88 and the degree that a required accuracy needs can be chosen beforehand.
 89 (5) *Where the medium is piecewise constant on the cells the projection error*
 90 *vanishes*, a constant suffices for the medium, and the degree of the field
 91 alone sets the order: in a randomly stratified model one first-moment voxel

92 per model layer reaches a reflection accuracy of 10^{-6} , where the uniform-
 93 field voxel needs about forty (§9), and in the uniform layer it allows voxels
 94 58 times larger in each direction (§11). (6) *In three dimensions* the smoothly
 95 graded sphere converges at second order with the mean-only voxel, as the
 96 layer predicts (§10), and at fourth order with the first-moment voxel in the
 97 companion paper [2].

98 *What makes the experiment exact.* The answer above can be trusted only if
 99 nothing but the two bases is approximated. Three results, set against the
 100 coupled-dipole method described below, secure that. (a) *An exact coupling.*
 101 The integrated Green tensor and the filtered coupled dipoles approximate the
 102 field of a cell; for cubes the source-cell average is exact, because the cube's
 103 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and
 104 at normal incidence the lattice kernel is in closed form with no lattice sum.
 105 (b) *No polarisability to choose.* The polarisability, a modelling choice in the
 106 coupled-dipole method, is here a derived object (§6): the moments of the
 107 Green's tensor over the cube, taken as distributions, yield Eshelby's δ term
 108 without adding it by hand, the cube's departure from the sphere as a pure
 109 cubic anisotropy, and the closure's error against exact scattering by a sphere
 110 as a real $(ka)^2$ term in closed form. The self term the single site carries is
 111 the one the averaged kernel omits, so the discrete system is a collocation of
 112 the continuum equation. (c) *The elastic state.* A voxel carries a force and
 113 a moment, so its state has nine components, the coupling is a 9×9 tensor,
 114 and its local term is the depolarisation of a plate.

115 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
 116 *space*, so a uniform region discretised into cubes is still that uniform region,

117 whereas spheres or point scatterers can only approximate it and carry a pack-
 118 ing error of their own. Their *regular lattice* makes the interaction between
 119 cells translation invariant, so the coupling is a convolution (Toeplitz between
 120 planes, Bloch-periodic within them), and the whole system is solved by fast
 121 transforms. And a cube small compared with the wavelength has a *single-site*
 122 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
 123 approximation a voxel introduces is the internal field it assumes.

124 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of
 125 small scatterers that interact through the background Green’s function is the
 126 multiple-scattering description of Foldy [3] and Lax [4]. In electromagnetics
 127 it became a numerical method as the coupled-dipole method, or discrete
 128 dipole approximation (DDA): each cell of a cubic grid is a polarisable point
 129 dipole driven by the incident field and by the fields of every other dipole
 130 [5, 6]. The method is a collocation of the volume integral equation for the
 131 electric field, with the cell’s self-interaction absorbed into its polarisability
 132 [7], and Yurkin and Hoekstra [8] review it in that form. Its accuracy turns on
 133 two ingredients, and this paper takes up both for elasticity. The first is the
 134 polarisability. The Clausius–Mossotti value is the static response of a sphere;
 135 the lattice dispersion relation of Draine and Goodman [9] adjusts it so that
 136 an infinite lattice propagates waves as the continuum does to a given order
 137 in kd ; Rahmani et al. [10] instead derive it from the exact static solution for
 138 the whole body, so that the method is exact in the long-wavelength limit.
 139 The second is the coupling. Integrating the Green tensor over the source
 140 cell rather than sampling it at the centre improves the accuracy, most of
 141 all for strong contrasts [11], and the filtered coupled dipoles of Piller and

142 Martin [12], revived by Yurkin et al. [13] for large refractive indices, replace
 143 the point dipole by a source of finite extent whose spectrum is band-limited
 144 to the grid. The coupling summed over a lattice extends the method to
 145 periodic structures [14]. Its error converges at second order in the dipole
 146 spacing when shape errors are absent [1], and for a cube, the one shape
 147 a cubic grid represents without a staircase, Yurkin and Kahnert [15] reach
 148 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. The
 149 layer studied here is the elastic counterpart of that cube: a body the grid
 150 represents exactly, so that the discretisation error can be isolated. What this
 151 paper adds is the closed form: for cells that tile space, the coupling that
 152 makes the lattice reproduce the continuum exactly, and the order and the
 153 constant of the error that remains, as functions of the two bases. In the
 154 terms used here the coupled-dipole method is the pair $p = r = 0$, and its
 155 refinements improve the polarisability and the coupling while leaving both
 156 bases constant in the cell.

157 *Its elastic counterparts.* The elastic problem has the same volume integral
 158 equation [16], and every cell-based solver of it is a coupled-dipole method
 159 whether or not it carries the name. Nestor [17] divided a two-and-a-half-
 160 dimensional medium into cells that interact through the background Green's
 161 tensor, each carrying a contrast renormalised by its own near field, with
 162 the multiple-scattering series summed. Kanaun and Levin [18] discretise the
 163 inclusion problem with Gaussian approximating functions whose matrix ele-
 164 ments are analytic. Yang et al. [19] solve a stress-velocity form of the integral
 165 equation on a grid by conjugate gradients with fast Fourier transforms, and
 166 Touhei [20] does the same for a half space with a generalised Fourier trans-

167 form. Jakobsen [21] and Jakobsen et al. [22] build the seismic forward and
 168 inverse problems on the transition operator of the discretised Lippmann–
 169 Schwinger equation, in the anisotropic elastic case with a nine-component
 170 state vector. The elastic problem adds a structure the scalar and electro-
 171 magnetic ones lack. A cubic voxel responds to the incident field with a force
 172 (through its density contrast) and a moment (through its stiffness contrast),
 173 so the state at a voxel has nine components (displacement and strain), and
 174 the single site’s self-interaction is the elastic depolarisation of a cube, the
 175 discrete counterpart of the inclusion tensor of Eshelby [23].

176 *This paper.* It extends the formulation of Nestor [17] to cubic voxels, and
 177 is ordered as the experiment is built. §2 sets out the discrete medium and
 178 §3 the continuous layer it must reproduce. §4 shows why the point coupling
 179 fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives
 180 the single-site T -matrix from moment integrals, tests it against the exact
 181 sphere, and shows that with it the discrete system is a collocation of the
 182 continuum equation. The wavefield basis comes next: §7 gives the second-
 183 order error of the uniform field in closed form, and §8 the voxel whose field
 184 carries Legendre moments to degree p , with its order $2p+2$ and its constant,
 185 for incident P and S at normal and oblique incidence. The medium basis
 186 follows in §9: a randomly stratified layer, for which a constant in each cell
 187 is exact, and a medium that varies within a cell, for which the contrast is
 188 expanded to degree r , the order becomes $\min(2p+2, 2r+2)$ and the error of
 189 the single-site T -matrix is the relative projection error of the medium. §10
 190 carries the prediction to a sphere, §11 discusses what the results imply for
 191 voxel schemes in general, and §12 closes with why an error known in advance

192 is worth having. Every result is derived symbolically in Mathematica and
 193 checked against an independent Python implementation or an exact solution.

194 2. Setting

195 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 196 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 197 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 198 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 199 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 200 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

201 displacement followed by engineering strain. A voxel responds with a force
 202 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 203 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 204 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

205 with ψ^0 the incident state.

206 *Where the kernel comes from.* Write the medium as the background plus
 207 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 208 $\rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Split-
 209 ting $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta\lambda \delta_{bc} \delta_{de} + \Delta\mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$
 210 the contrast stiffness, and moving the contrast terms to the right leaves the
 211 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta\rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

212 a force from the density contrast, and the divergence of the contrast stress m ,
 213 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the
 214 background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term
 215 integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

216 the boundary term vanishing because m is confined to the scatterers, and the
 217 last step holding because $\mathbf{G}$ depends only on the separation. The moment
 218 therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when
 219 the moment is the contrast stress itself. With the moment tensor defined
 220 instead through its equivalent body force $-\partial_c[M_{bc}\delta(\mathbf{x}-\mathbf{y})]$, as in Aki and
 221 Richards [24], the same field reads $M_{bc}\partial G_{ab}/\partial y_c$, the source derivative, with
 222 $M = -m$. The two are one convention written two ways; what matters is that
 223 the derivative in the kernel and the sign in the contrast operator are chosen
 224 together. The nine-component kernel follows (Fig. 1): its force columns are
 225 $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over
 226 (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are
 227 those entries and its strain rows their receiver derivatives, symmetrised in
 228 the same way and doubled on the shear rows to give engineering strain. In
 229 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

230 where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$
 231 and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

232 *Averaging, and collocation.* Three operations in this paper are averages over
 233 a cell, and they are distinct. The *source-cell average* integrates the kernel
 234 over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ is the field at $\mathbf{r}$
 235 of a uniform polarisation of the source cell. Every scheme in this paper uses
 236 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 237 is imposed: at the receiving cell's centre, with the internal field of each cell
 238 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 239 *average* imposes the equation in the mean over the receiving cell instead;
 240 with a uniform internal field that is the Galerkin scheme of degree zero,
 241 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 242 term that refers to the internal field and not to the test, and they differ only
 243 in the test function (a delta at the centre against the cell's indicator) and
 244 in the constant of their $O((kd)^2)$ error (Table 3). The *cell-mean contrast*,
 245 finally, is an average of the material, not of the field: where the medium
 246 varies within a cell, the contrast is projected onto the cell's basis, its mean
 247 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 248 of the coupled-dipole method, the point dipole is collocation with the point
 249 kernel, the integrated Green tensor adds the source-cell average, and the
 250 filtered coupled dipoles replace the point source by one of finite extent.

251 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 252 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 253 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 254 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes

255 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

256 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 257 join the centres of the voxels in one plane (a square lattice of pitch d), and
 258 m is the separation of the two planes in units of d (Fig. 2a,b). The question
 259 is whether, and how fast, this chain converges to the continuous layer as
 260 $n \rightarrow \infty$ at fixed D .

261 *Implementations.* Every result is derived symbolically in Mathematica and
 262 compared with an independent numerical implementation or an exact solu-
 263 tion. “The package” below is the Python implementation of the voxel scheme
 264 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 265 propagator, which serve both as the object under test and, where validated,
 266 as a second implementation.

267 3. The continuous layer and its thin-layer expansion

268 At normal incidence a plane force drives a one-dimensional problem per
 269 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 270 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 271 follow from continuity of displacement and traction at its faces; they agree
 272 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 273 P and S (`ContinuumLimit_Reference.wl`).

274 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 275 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their

ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coefficient expands as

$$R = \frac{ik_1 D}{2}(\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4}(\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24}(\zeta - \zeta^{-1})(3\zeta^{-2} + 2 + 3\zeta^2) + \frac{(k_1 D)^4}{48}(\zeta^2 - \zeta^{-2})(3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (7)$$

exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each coefficient symbolically and the truncated series against the exact layer). The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

to first order a thin layer is an interface across which the traction jumps by the excess inertia and the displacement by the excess compliance. At oblique incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$ for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target a single plane of voxels must reproduce order by order. A plane of voxels that carry Legendre moments of their internal field to degree p reproduces Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the mean-only voxel reproduces the equivalent interface and the D^2 term, the first-moment voxel all four terms shown.

294 4. The specular sums of the point Green's tensor

295 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 296 of every voxel of a plane m planes away. In that form it is a poor object to
 297 study. The Green's tensor decays slowly with distance, so the sum converges
 298 slowly, and it hides the one question that matters here: which part of it is
 299 the continuum, and which part is the discretisation. The Poisson summation
 300 formula [25] answers both at once, and is the standard tool for lattice sums
 301 of wave fields [26]. For a function f on the plane and the square lattice of
 302 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

303 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 304 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3).
 305 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

306 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 307 between planes a distance z apart. Each term has a physical meaning. Trans-
 308 formed laterally, the field of a plane of sources is a superposition of plane
 309 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 310 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 311 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 312 far beyond the wavenumber of the field when the voxels are small compared
 313 with the wavelength, so it is evanescent and decays away from the plane as

314 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 315 the lattice sum exactly into the continuum and a discretisation correction,
 316 and the correction converges after a few orders.

317 *The local term that survives.* In the strain-from-moment block the evanescent
 318 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 319 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 320 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 321 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 322 planes in slab order, the static lattice sum of the point kernel is exactly
 323 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 324 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 325 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 326 reproduces the continuum only if that remainder is cancelled by the voxel's
 327 own self term.

328 5. The tiling identity

329 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r}-$
 330 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 331 Each Poisson order in Eq. (10) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

332 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 333 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 334 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 4).
 335 Two exact statements follow.

336 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

337 the continuum's plane-wave term times the vertical form factor. The pack-
 338 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 339 `AveragedSums.wl`).

340 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 341 is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

342 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 343 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 344 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 345 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 346 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 347 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 348 with every cell integral evaluated analytically (below).

349 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 350 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

351 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 352 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 353 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}

at its default settings, falling to 2.6×10^{-5} only when both the shell radius and the quadrature order are raised; Eq. (14) is now the package's kernel at normal incidence.

Cell integrals without quadrature. Write the static (Kelvin) tensor as $G_{bc} = (c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$. Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$, which never contain the origin, and there the two-dimensional antiderivative in the face coordinates is continuous, so the face integral is a sum over the face's corners. The distributional term at $r = 0$ is carried by the divergence theorem itself. (A triple antiderivative evaluated at the cube's corners is *not* usable for repeated derivatives: its arctangent terms jump across the coordinate planes through the origin, and differentiating twice there discards exactly that term.) The self term so obtained reproduces the classical closed form of the cube's depolarisation to machine precision.

6. The single site: the cube T -matrix from moment integrals

Every error of a voxel scheme is made at the single site (§11), so the single site is derived here rather than imported. The classical route imports it. For an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a uniform load uniform, with a depolarisation tensor in closed form [23, 27, 28]; its dynamic generalisation is known for spheres and ellipsoids [29, 30]. For a cube or a general polyhedron the static field of a uniform eigenstrain is also in closed form [31, 32, 33, 34], but the strain it produces is not uniform, so there is no exact uniform-strain inclusion to import. In scattering theory

378 the same object appears as the self-consistent or effective-field closure of
 379 the volume integral equation [35, 16, 36]: a first-order (Born) scatterer [37]
 380 renormalised by the geometric series of its own near field, the step that turns
 381 a contrast into an effective contrast. Nestor [17] introduced the single site
 382 in this form, summing the series of the cell's self integral, with the singular
 383 near field evaluated over an infinitesimal cell. The moments below evaluate
 384 the same self integral over the finite cube, as a distribution. The elastic T -
 385 matrix of Waterman [38] represents a scatterer in spherical multipoles, the
 386 natural basis for a sphere; a voxel in a lattice needs one compatible with the
 387 lattice, and the moments over the cube provide it. The construction below
 388 makes the approximation of a voxel explicit rather than implicit: it derives
 389 the cube's T -matrix from the moments of the elastodynamic Green's tensor
 390 over the cube, and identifies the uniform-strain closure as one truncation of
 391 an exact hierarchy.

392 6.1. *The closed hierarchy*

393 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda\delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 394 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 395 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 396 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed

397 system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (16)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

398 all fields evaluated at the centre. Its coefficients are six moments of the
399 background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \end{aligned} \quad (18)$$

400 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
401 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
402 it, so it is fixed by convention. The moments are graded by (D, W) , the
403 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
404 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

405 6.2. The moments as distributions

406 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (19)$$

407 [39, 24]. Expanding g_c in powers of r reduces every moment in Eq. (18) to a
408 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (20)$$

409 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
410 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
411 gral depends on the shape of the region excluded around the origin. This is
412 the source of the δ -function term of the inclusion problem, which a numerical
413 quadrature misses. Without it the depolarisation is wrong; for a sphere it
414 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
415 are distributional from the start, and each is moved onto the voxel's faces by
416 pairing with the test function $w\mathbf{1}_V$, using $\partial_q\mathbf{1}_V = -n_q\delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (21)$$

417 The surface term is evaluated on the faces, a distance h from the origin, where
418 every derivative of r^m is smooth. The recursion lowers D and W together
419 and ends on convergent integrals. The face integrals are done in closed form
420 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (22)$$

421 which the face route satisfies exactly and a spherical excision returns as
422 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell
423 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
424 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
425 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
426 Because every power of r is paired this way, the dynamic moments keep the
427 even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones.
428 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
429 those terms, at a real $O((kh)^2)$ cost (§6.4).

430 6.3. Parity, and the nine-component block

431 The cube is centrosymmetric. A moment that couples a field quantity of
 432 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 433 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (16) contains the first gradient alone: ∂u is
 434 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 435 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 436 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 437 by Eq. (16), and *no treatment of the second-gradient block (17) can change*
 438 *it*. What Eq. (17) contributes reaches the far field only through the $O((kh)^2)$
 439 coupling in Eq. (15). That is the order, and the origin (the variation of the
 440 field across the voxel), of the collocation error of §7, which the first-moment
 441 voxel of §8 removes.

442 With $\Delta\rho = 0$, Eq. (16) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 443 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 444 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 445 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (23)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (24)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (25)$$

446 The density channel is the displacement block of Eq. (15) with the second
 447 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 448 of $\mathbf{G}$.

449 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 450 tional channel $1 + \Delta K / (\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 451 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 452 latational channel, Eq. (23), is the sphere's exactly: its eigenvalue involves
 453 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 454 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 455 the shape. The cube's shear channels differ from each other. Their isotropic
 456 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (26)$$

457 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 458 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 459 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 460 and the diagonal shear is the more depolarised. At the background of §7
 461 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 462 0.4960.

463 *Where Eshelby enters.* Equations (23)–(25) follow from the moment integrals
 464 and the group theory alone. No ellipsoidal result is used, and no δ term is
 465 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 466 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 467 the uniformity theorem that ansatz is exact if and only if the inclusion is
 468 an ellipsoid. For a sphere the construction is therefore exact in the static
 469 limit, and for a cube it is an approximation. The hierarchy (15)–(17) states
 470 that approximation as the omitted blocks, rather than as an unquantified
 471 shape correction. A polynomial expansion about the centre cannot represent

472 the edge and corner concentrations of an isolated cube's internal field at any
 473 finite degree. A voxel in a uniform layer has no such concentrations: its
 474 neighbours have the same properties, so its internal field is the continuum's,
 475 which is smooth. That is why a low-degree internal field suffices in a lattice
 476 (§§7–8) although it is only an approximation for an isolated cube.

477 *Against the exact sphere.* These statements are tested against the exact elas-
 478 tic Mie solution for a sphere, with no Eshelby formula in between. The
 479 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
 480 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
 481 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
 482 route of Eq. (21) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
 483 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
 484 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 485 isotropic shear average, Eq. (26), is the depolarisation read from Mie's E_2
 486 to 10^{-8} . The same moment taken as a principal value, over the ball with
 487 a small sphere excluded around the origin, has zero depolarisation in every
 488 channel. The whole static response of the sphere is the δ term.

489 At finite frequency the closure uses the dynamic ball moments, which are
 490 in closed form. It departs from Mie at second order in ka and first order in
 491 the contrast, and the departure is closed-form too: the ball's moments are
 492 elementary, and Mie's low-frequency series with the contrast kept exact gives

493 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (27)$$

494 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 495 the radiation damping, agree exactly in all three channels. So the closure's
 496 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 497 own. That is the order the hierarchy assigns to the omitted second gradi-
 498 ents (§6.3), and it has the same origin, the variation of the field across the
 499 scatterer, as the collocation error of §7. Eq. (27) is derived symbolically
 500 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 501 the numerical Mie solve and an independent implementation of the closure
 502 to 5×10^{-4} , which is the size of the next order in the contrast.

503 6.4. The collocation closure

504 The collocation closure takes the internal field of a voxel as uniform, the
 505 truncation of §6.3 for the density and stiffness channels together. The voxel's
 506 own field at its centre is then $\mathbf{S}_{\text{self}} \Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 507 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 508 So $e = \psi + \mathbf{S}_{\text{self}} \Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (28)$$

509 With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 510 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (29)$$

511 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 512 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
 513 so Eq. (29) is precisely the collocation of the continuum Lippmann–Schwinger
 514 equation with a strain piecewise constant over voxels. It does not contain
 515 $\mathbf{S}_{\text{self}}$ at all.

516 The package’s cube T -matrix is this closure statically: both blocks agree
 517 with Eq. (28) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 518 (`ContinuumLimit_CubeT.wl`). It departs dynamically by about $0.25(kh)^2$
 519 through its far-field form-factor corrections. Its self integrals are exact in the
 520 static and radiation parts but omit the real even-power terms of the Green’s
 521 tensor’s expansion ($k^2 r, r^3, \dots$), an error of $0.4\text{--}0.6(kh)^2$ against the analytic
 522 self term, whose series agrees with the closed-form elastodynamic tensor to
 523 10^{-14} . By Eq. (29) that error cannot enter the closure scheme.

524 7. Convergence, and the closed-form error

525 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
 526 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
 527 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 528 the chain converges to the continuous layer’s at order 2.000 for $n = 4 \rightarrow$
 529 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (29) and with the
 530 package T -matrix alike (`ContinuumLimit_Chain.wl`). The error depends on

531 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
 532 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
 533 the exact solution and n the refinement; the orders in the legend of Fig. 5
 534 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
 535 with their standard errors. The two agree where a sequence is asymptotic
 536 and differ where it is not, which the apparent orders show.

537 *The leading error in closed form.* In Eq. (29) the kernel is integrated exactly;
 538 the only approximation is the internal field's variation across a voxel. At first
 539 order in the contrast the discrete scattered field is therefore a midpoint rule
 540 applied to the continuum's Born integral, with the kernel's cell integral exact.
 541 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 542 it is flat. With k the background wavenumber of the wave concerned (k_P for
 543 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 544 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (30)$$

545 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 546 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 547 and those of Eq. (33) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 548 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 549 the transmission constant grows to -0.092 .

550 *Why a uniform-field single site stops at second order.* The two constants have
 551 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 552 cancel both. The package *T*-matrix illustrates it: its form-factor corrections

553 reduce the reflection error by a factor 4.45 and *increase* the transmission
 554 error by 1.7, at every n . The missing term in Eq. (30) is the first moment
 555 of the internal field over each voxel, which radiates through the derivative of
 556 the kernel. A fourth-order scheme must carry it: a single site with a gradient
 557 channel, and a kernel whose cell integrals include the kernel's derivative.

558 8. Fourth order and beyond: Legendre moments of the voxel field

559 Let each voxel carry the mean *and* the first moment of its internal field.
 560 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) =$
 561 $P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w =$
 562 $\sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the
 563 polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the
 564 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (31)$$

565 with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (32)$$

566 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 567 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$
 568 and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form.
 569 Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 con-
 570 verges at fourth order in both reflection and transmission (`ContinuumLimit_`

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (32). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

571 **FourthOrder.wl**; Table 1). A single plane of first-moment voxels represents
572 the 2 m layer 6500 times more accurately than a single plane of uniform ones.
573 The collocation column is the scheme of §6 and reproduces the chain of the
574 full 9×9 system to all printed digits.

575 *Higher moments.* Carrying the Legendre moments to degree p changes noth-
576 ing in the construction: the moments and the same-cell double integrals of
577 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-
578 degree solver reproduces the chain above exactly. The second-moment voxel
579 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
580 eighth, in reflection and transmission (Table 2; **ContinuumLimit_SecondMoment.**
581 **wl**); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
582 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

583 *The leading error in closed form.* Without scattering, the Galerkin solution
584 is the L^2 projection Π of the incident field onto each voxel's basis, and the
585 kernel is integrated over each voxel exactly; at first order in the contrast the

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} , $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (33)$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 3. The chain of Eq. (32) reproduces the forms for collocation and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7} at a contrast reduced by 10^{-6} . The constant for degree p follows from the Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n / (2n+1)!!$ and

Table 3: The leading error of the discrete layer in closed form, Eq. (33): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

600 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (34)$$

601 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
602 mission for odd p , opposite signs for even p . The first-moment voxel's leading
603 error therefore does not depend on the direction in which the field is observed.
604 Eq. (34) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
605 ishes) in Mathematica and, independently, in sympy.

606 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
607 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
608 $d = D$) the series in D of the reflection coefficient, computed exactly at the
609 full contrast, coincides with that of the exact layer, Eq. (7), term by term
610 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
611 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

612 `w1`). Each further Legendre moment adds two terms of the thin-layer series.

613 *In three dimensions.* The three-dimensional first-moment voxel carries the
 614 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 615 in all, tested with the same functions and coupled by cell-to-cell double in-
 616 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 617 cidence it reduces *exactly* to Eq. (32) (`ContinuumLimit_Reduction3D.w1`).
 618 The lateral first moments are never excited: the lattice and the incident
 619 field are symmetric under reflection about every voxel's centre planes, and
 620 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 621 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 622 ishes on the reciprocal lattice and the plane sum of the double integrals
 623 is the plate-to-plate integral in z alone. Two facts mark where genuinely
 624 three-dimensional structure begins. The lateral first moment's form fac-
 625 tor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the recipro-
 626 cal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the
 627 evanescent orders. And at oblique incidence the mean's own zeros shift,
 628 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds
 629 only to $O(k_{\parallel} h)$.

630 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
 631 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g},$$

(35)

632 with L_a the lateral Legendre factors. The whole-space tensor decays as
 633 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each

entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[i\alpha/2\gamma + i\beta \text{sgn}(z)/2] e^{i\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (35). At 40° the orders are again 2 and 4.

	mean only		mean and first moments	
n	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (32) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (32) with $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full 9×9 construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique inci-

649 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

650

651 of Eqs. (30) and (33), are those of the incident P wave.

652 *Degree two at oblique incidence.* In three dimensions the voxel of degree 2
653 carries, on each of the nine components of its state, the ten Legendre products
654 of total degree at most two in (z, x, y) . In the construction of Eq. (35) it
655 converges at sixth order against the exact layer for incident P, SV and SH
656 waves (Table 6; `scripts/measure_oblique_degree.py`). The test is run at
657 $\omega = 3000 \text{ rad s}^{-1}$ ($k_P D = 1.2$, $k_S D = 2$), ten times the frequency of the other
658 tables, because at 300 rad s^{-1} a sixth-order error on these grids lies below
659 the round-off of double precision. The exact reflection coefficient is that of
660 Kennett’s recursion, an independent algorithm, and the same computation
661 with the bases of degree 0 and 1 returns orders 2 and 4 in all five cases.
662 At normal incidence the error on four planes, 6.95×10^{-9} , is the $c_2(kd)^6 =$
663 7.2×10^{-9} of Eq. (34) to 4%. Doubling the number of lattice orders retained
664 in Eq. (35) changes the errors by less than 1%. Unlike the degrees 0 and 1,
665 this degree is implemented once at oblique incidence; its check is the exact

solution.

Table 6: The voxel of degree 2 (ten Legendre products per state component) at $\omega = 3000 \text{ rad s}^{-1}$: relative error of the same-type reflection coefficient against the exact layer on n planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

incident wave	error, $p = 2$			order	orders
	$n = 1$	$n = 2$	$n = 4$	$p = 2$	$p = 0$ / $p = 1$
P, normal: R_{PP}	3.4×10^{-5}	4.6×10^{-7}	7.0×10^{-9}	6.05	2.04 / 4.05
P at 20° : R_{PP}	7.3×10^{-6}	1.0×10^{-7}	1.5×10^{-9}	6.04	2.04 / 4.06
SV at 20° : R_{SS}	2.2×10^{-4}	2.3×10^{-6}	3.4×10^{-8}	6.11	2.10 / 4.14
SH at 20° : R_{SH}	2.3×10^{-4}	2.5×10^{-6}	3.6×10^{-8}	6.11	2.10 / 4.14
SH at 40° : R_{SH}	3.0×10^{-4}	3.6×10^{-6}	5.2×10^{-8}	6.09	2.12 / 4.05

666

667 The spectral construction is implemented twice. The second implementa-
668 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
669 nition of the scheme: it takes the depth dependence of the transformed kernel
670 from an independent residue construction, obtains the local weight as a nu-
671 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
672 ble integral by quadrature and separates P from S by sampling the upgoing
673 field at two depths. The raw reflected displacements of the two agree to $9 \times$
674 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
675 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
676 scheme of any degree is also implemented twice. The second implementation
677 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
678 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
679 from its own interface system, in double precision. Its errors agree with the

680 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
 681 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 682 6.00; for one plane as the layer thins its differences from the exact layer fall
 683 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 684 Figure 5(a,b) shows the convergence, with the orders fitted to it.

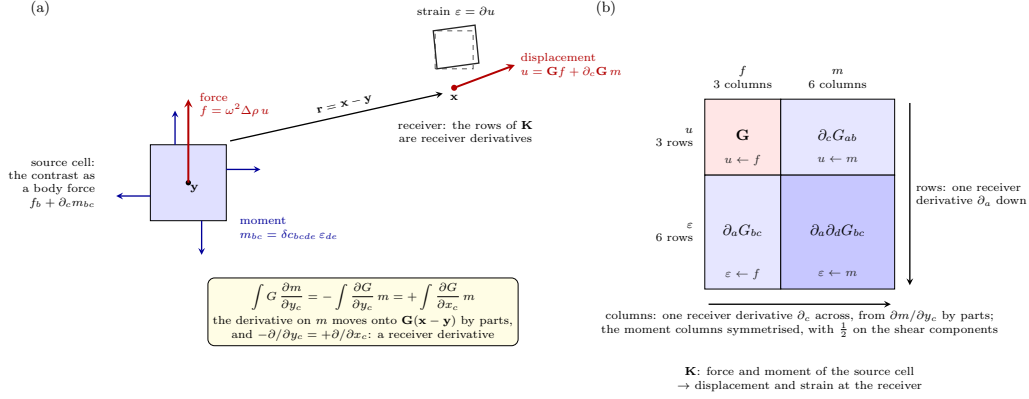


Figure 1: Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m , the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε . The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled u , three rows, and strain, six rows, and columns labelled f , three columns, and m , six columns; the blocks read $\mathbf{G}$, the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

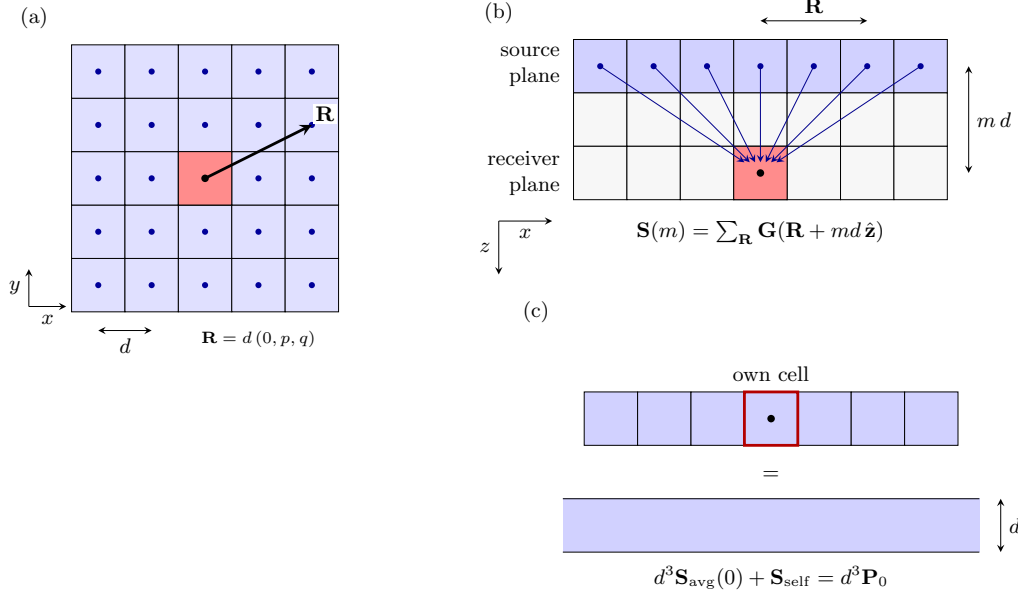


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (shaded darker) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

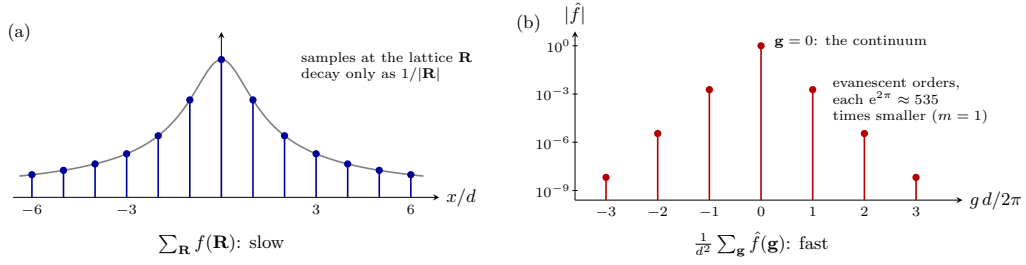


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

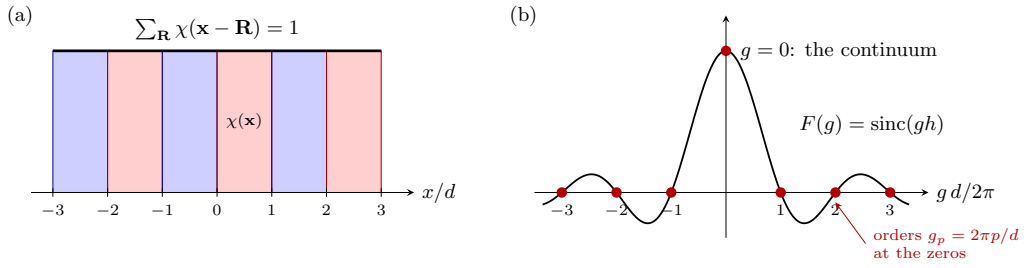


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

685 9. A heterogeneous layer

686 A heterogeneous medium is not a layer of identical cells. The simplest
687 heterogeneous case with an exact answer is a stratified model built from layers
688 of constant properties, with real interfaces between them: the form of many
689 layered crustal velocity models. The last test takes such a model. A layer
690 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
691 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
692 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
693 stratification with physical contrasts. The voxels refine the model's layering,
694 with m planes of voxels in each model layer, so no voxel straddles a model
695 interface: the natural discretisation when the model's interfaces are known.
696 At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
697 reverberates between the model interfaces. The exact answer is the product
698 of the model layers' propagators. Its same-type reflection coefficients agree
699 with those of the reflection–transmission recursion of Kennett [40], computed
700 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
701 now changing from plane to plane; nothing else changes. The errors are in
702 Table 7 and Fig. 5(c).

703 *The interfaces cost nothing.* Every order is the one measured for the uniform
704 layer. At a model interface the strain of the true field jumps. A voxel basis
705 that is discontinuous from voxel to voxel represents that jump exactly when
706 the interface lies on a voxel boundary, so the interface adds no error of its
707 own. The only requirement is that voxel boundaries coincide with the model's
708 interfaces. A voxel that straddled an interface would carry a contrast that is
709 not uniform within it, a case not examined here.

Table 7: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

710 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
711 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
712 layer. The first-moment voxel needs 8, one per model layer: the model's own
713 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
714 against 36), that is 3123 against 288 per vertical column of voxels, a factor
715 of eleven. In three dimensions, with the lateral grid refined as the vertical
716 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
717 unknowns by about 2×10^4 . The same model has been solved independently
718 in Python, with a contrast for each plane: the raw reflected displacements
719 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
720 at 20° , with and without the first moments. Lateral heterogeneity needs the
721 full three-dimensional solver; it is the first extension discussed below.

722 *A medium that varies within a cell.* The stratified model is exact on its own
 723 grid: every cell is homogeneous. A smoothly varying medium is not, and
 724 sampling it as cells of constant properties is an approximation of its own.
 725 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
 726 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
 727 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
 728 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (32)
 729 becomes

$$\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (36)$$

730 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 731 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 732 the order of convergence is

$$\min(2p+2, 2r+2), \quad (37)$$

733 or $2p+2$ when degree r represents the profile exactly (Table 8; `ContinuumLimit_`
 734 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 735 caps every voxel at second order, however rich its field basis, and for the
 736 smooth profile the voxel of degree 2 with a linear contrast has the error
 737 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 738 this is what one expects: the contrast's projection error is orthogonal to
 739 the polynomials of degree r , so against the smooth product of the inci-
 740 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 741 the contrast must be represented to the degree of the field. The closed forms
 742 of the construction are tested one by one by an independent implementa-

Table 8: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (37), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
bles the scheme without linearisation, evaluating the projected contrast at
the quadrature nodes, against its own 30-digit solution of the graded layer;
its errors agree with those of Table 8 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it
measures the orders of Eq. (37) again.

Which basis sets the error. Equation (37) gives the order. The size of the
error, and which of the two bases it belongs to, appear when the scattered
field is separated by its order in the contrast: the Born term T_1 , and the
second-order term T_2 in which the field scattered once is scattered again.
Both follow from the solution at contrasts scaled by $\pm\epsilon$ and $\pm 2\epsilon$, for the

756 scheme and for the exact layer alike. With $f(z)$ the profile and Π_q the L^2
 757 projection onto the Legendre polynomials of degree at most q in each cell,
 758 define the *relative projection error* of the medium at degree q ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (38)$$

759 the fraction of the squared norm of the profile that polynomials of degree q
 760 on the cells cannot represent. It is a property of the medium and the grid
 761 alone; it involves neither the wave nor the solution. For the smooth profile
 762 and every one of the nine pairs with $p, r \leq 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (39)$$

763 to within 1.3% at $\omega = 300 \text{ rad s}^{-1}$ on 4 to 16 cells: the ratio of the two sides
 764 is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2.
 765 The error of the Born term is smaller by two to six orders of magnitude. The
 766 error of the whole is therefore that of T_2 , the projection error times $|T_2|/|T_1|$
 767 (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7} , and Table 8
 768 has 1.4×10^{-7} .

769 Equation (39) is exact in the long-wave limit, for a reason that shows what
 770 each basis is for. The strain-strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$,
 771 is local as $kD \rightarrow 0$, and the incident strain ε_0 is then uniform across the layer,
 772 so the strain scattered once is $-(\Delta M/M_P) f(z) \varepsilon_0$: it is the profile itself. A
 773 cell holds the contrast as $\Pi_r f$ and the field to degree p , so it holds the once-
 774 scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order
 775 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) dz = \|\Pi_q f\|^2, \quad (40)$$

776 the last because $\Pi_q f$ lies in the range of Π_r . The relative error is $1 -$
 777 $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$. Because the kernel is local in this limit, the second
 778 scattering takes place in the cell of the first: T_2 is the second-order term of
 779 the single-site T -matrix, and Eq. (39) is a statement about the single site.
 780 The T -matrix sums the rescattering within a cell exactly for the cell as it
 781 is represented, a contrast $\Pi_r f$ and a field of degree p ; its error is that the
 782 cell is not the medium. The split is measured directly (`scripts/measure_`
 783 `layer_bases.py`): at $\omega = 300 \text{ rad s}^{-1}$ on 4, 8 and 16 cells, T_2 of the scheme
 784 and of the exact layer are each divided into the part whose two scatterings
 785 lie in one cell and the part between different cells. For all nine pairs the
 786 error of the same-cell part is 0.998 to 1.013 times $|T_2| \mathcal{E}_{\min(p,r)}$, and that of
 787 the part between cells at most 0.002 times it. The coupling between cells
 788 contributes only the departure from the limit. Measured, the ratio of the
 789 two sides of Eq. (39) for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2} ,
 790 4.3×10^{-3} , 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24, 0.12, 0.04$ and 0.012 : as
 791 $(kD)^2$ (`scripts/measure_layer_bases.py`). The pairing in Eq. (40) is ver-
 792 ified symbolically for all nine pairs, and the projection errors are computed
 793 exactly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each halv-
 794 ing of the cells.

795 Three consequences follow. The two bases must be matched: a contrast
 796 of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$
 797 have the errors of $(0, 0)$ to three digits), and a field of higher degree than
 798 the contrast improves only the Born term. The wavefield basis is there to
 799 carry what the medium imprints on the field, not only the wave: at these
 800 frequencies the wave itself varies little across a cell. And the error is known

801 from the medium, before anything is solved: $\mathcal{E}_q$ is a property of the profile
802 and the cells. It vanishes when the medium is piecewise constant on the cells,
803 which is the stratified model above; there the contrast needs no more than a
804 constant in each cell, and the degree of the field alone sets the order.

805 10. A sphere: what the layer predicts in three dimensions

806 The layer and the stratified layer are laterally uniform. A sphere is not,
807 and elastic scattering by a sphere, homogeneous [41, 42] or radially varying
808 [43], has an exact solution: it is the usual benchmark for volume-integral and
809 discrete-dipole methods. This section solves two spheres with the full three-
810 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
811 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
812 iterative solver, and a P wave is incident. For the graded sphere the voxel is
813 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
814 nine-component state, tested with the same constant, and the L^2 projection
815 of its contrast, which is its cell mean; every cell that overlaps the sphere is
816 kept, and the cells are coupled by cell-to-cell double integrals of the Green's
817 tensor, one block per orbit of the cube group [2]. For the sharp sphere
818 the voxel is the package's cube T -matrix, which differs from the collocation
819 closure only through its far-field corrections (§6.4) and has the same order,
820 each cell keeping the contrast of its centre. The far field is computed from
821 the solution for the incident plane wave itself, each voxel's displacement and
822 strain carrying the plane-wave phase at its centre; adding to that phase a
823 linear variation about the scatterer's centre, as a Taylor expansion of the
824 incident field would, counts the wave's variation twice and leaves an error of

825 order $(ka)^2$ that no refinement removes. The error is the largest difference
 826 between the scattered far field and the exact one over nine scattering angles
 827 from 11° to 169° , relative to the largest exact value, at a distance of 5×10^8
 828 radii for the graded sphere: at 5×10^4 radii the near-field terms that the
 829 asymptotic far field omits are 1.4×10^{-4} of the peak, 4% of the discretisation
 830 error at 12 voxels across and most of it at 24. The sharp sphere, whose errors
 831 are a hundred times larger, is observed at 5×10^4 radii.

832 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 833 its shape with every grid. A sphere whose contrast falls smoothly to zero at
 834 its surface does not: the voxels that cross the surface carry a vanishing con-
 835 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 836 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 837 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 838 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 839 the radial equations of Takeuchi and Saito [43], integrated across the shell
 840 from the core's regular solutions and matched to outgoing waves at $r = a$.
 841 The equations are derived symbolically from the equations of motion, and
 842 the solution is checked against a sphere of homogeneous concentric shells
 843 in closed form, by unitarity and reciprocity, and by an independent imple-
 844 mentation that agrees to 2×10^{-11} (TakeuchiSaito.wl, ContinuumLimit_
 845 GradedSphere.wl, scripts/crosscheck_graded_sphere.py). Each voxel
 846 carries the mean of the contrast over its cell. Table 9 shows the error falling
 847 at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (scripts/pilot_
 848 graded_voxel_sphere.py): the apparent order between successive grids is
 849 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders

850 of Fig. 5(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across the
851 error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
852 graded shell and the order is not yet asymptotic. This is the order the layer
853 predicts for the mean-only voxel, and the first test of that prediction on a
854 body with genuine three-dimensional structure.

855 What makes the sequence this clean is that the voxel carries a single error.
856 The collocation voxel of the sharp sphere applied to this body, each cell at
857 the profile's value at its centre and only the cells whose centres lie inside the
858 sphere kept, carries two: its own internal-field error and the error of sampling
859 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
860 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
861 2.0, 3.5 and 3.6 at $k_S a = 0.5$, and at $k_S a = 1$ the apparent orders are 1.7, 2.5
862 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
863 but its apparent order is not that of any single term and cannot be read from
864 them. The same collocation voxel with the cell mean of the contrast, and
865 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
866 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
867 second term, and with it removed the collocation and mean-only voxels differ
868 only by their constants, as in Table 3. The cell mean, with every overlapping
869 cell kept, represents the same body on every grid, and the order the layer
870 predicts is then the one observed.

871 The first-moment voxel of §8, extended to three dimensions with a con-
872 trast linear in each cell, converges at fourth order on this same sphere [2]: its
873 apparent orders are 4.01 and 4.13 where the discretisation error is isolated
874 at weak contrast, and at the full contrast its error with 16 voxels across is

875 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below the mean-only voxel's on
876 the same grid.

Table 9: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{Sa} = 0.5$		$k_{Sa} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

877 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
878 and its voxelisation changes with every grid. Table 10 shows its error for
879 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_{Sa} = 0.5$.
880 The error tracks the staircase's volume, which is 5%, 17% and 4% too large
881 on these grids. Rescaled by the ratio of the sphere's volume to the staircase's,
882 it falls below 1% at $k_{Sa} = 0.5$ and to at most 3% at $k_{Sa} = 1.0$, but it still
883 does not converge: what remains is the staircase's shape. A refinement study
884 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
885 measures the scheme.

Table 10: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

886 11. Discussion

887 Figure 5 collects the convergence of every test, with the order fitted to
888 each by least squares (`scripts/plot_convergence_orders.py`).

889 *What each basis is for.* A voxel holds two functions, and the results give
890 each basis its part. The basis of the wavefield sets how well the voxel carries
891 the field across itself: in a uniform medium the scheme converges at order
892 $2p + 2$ with the constant c_p of Eq. (34), for P and S waves and at normal and
893 oblique incidence alike. The basis of the medium sets how well the voxel *is*
894 the medium: with degree r the order is capped at $2r + 2$ (Eq. (37)). Neither
895 can make up for the other, and the second-order term shows why. The field
896 scattered once is a copy of the medium’s profile, so the wavefield basis must
897 carry the profile as well as the wave, and the error of that term is the relative
898 projection error of the profile at the lower of the two degrees (Eq. (39)). At
899 the frequencies of interest the wave varies little across a cell and the medium
900 may vary much, so it is the medium, not the wavelength, that decides the

901 degree and the size of the cells. Two cases bracket the choice. Where the
 902 medium is piecewise constant on the cells, as in a layered model with voxel
 903 faces on its interfaces, a constant is exact for the medium, the projection error
 904 vanishes, and the degree of the field is spent entirely on the wave. Where the
 905 medium varies smoothly within the cells, the two degrees should be equal:
 906 a higher degree for the contrast than for the field is wasted, and a higher
 907 degree for the field improves only the Born term.

908 *Where the error of a voxel scheme lives.* The results separate the two ap-
 909 proximations a voxel scheme makes. The *coupling* between voxels makes
 910 none: averaged over the source cell it is the continuum's cell integral exactly,
 911 between planes by the form factor's zeros and within a plane by the tiling
 912 identity, and the self term it omits is exactly the one the single site carries.
 913 Every error therefore comes from the *single site*: from the internal field it
 914 assumes and, where the medium varies within the cell, from the medium it
 915 is given. Its T -matrix sums the rescattering within the voxel exactly, but
 916 for the voxel as represented, and the split of the second-order term in §9
 917 measures the consequence: the error sits in the part whose two scatterings
 918 lie in one cell, and the part between cells carries at most 0.2% of it. For a
 919 uniform field that error is the midpoint rule's, known in closed form; for a
 920 linear field it is the next term of the same expansion. The converse is the
 921 practical warning. A kernel that is not the cell integral (the point Green's
 922 tensor, or a cell average that does not match the single site's closure) leaves
 923 a local term that does not vanish as the voxels shrink (§4): the evanescent
 924 lattice orders carry a static contribution of fixed size per voxel volume. A
 925 scheme built that way converges to the wrong limit, and refinement cannot

926 reveal it, because the error is scale-invariant.

927 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
928 single site from a modelling choice into a derived object with a stated error.
929 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
930 only by the cubic split of Eq. (26). Its dynamic part departs from exact
931 scattering by a sphere only through the real $(ka)^2$ term of Eq. (27); the
932 radiation damping is exact. That term and the collocation error of Eq. (30)
933 have the same origin, the variation of the field across the scatterer, and the
934 same remedy, a richer internal field. In a lattice the remedy is cheap: the
935 field a voxel must represent is the smooth continuum field, not the field of
936 an isolated cube, with its concentrations at the edges and corners.

937 *Relation to the coupled-dipole method.* The same partition appears in the
938 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
939 relation, the lattice dispersion relation [9], the polarisability taken from the
940 exact static solution [10] and the integrated Green tensor [11] all adjust
941 the single site so that the lattice reproduces the continuum. For space-
942 filling cubes the adjustment is neither needed nor a free parameter: the tiling
943 identity fixes the local field exactly, the self term of the collocation closure is
944 the one that makes it cancel, and what is left is a closed-form error rather than
945 an empirical one. That the method is a collocation of the volume integral
946 equation was shown for the electromagnetic case by Lakhtakia [7]; Eq. (29)
947 is the same statement for the elastic lattice with the self term evaluated over
948 the finite cube, and it is what makes the error a midpoint-rule error. The
949 filtered coupled dipoles of Piller and Martin [12] and Yurkin et al. [13] band-
950 limit the source to the grid; the source-cell average does the complementary

951 thing exactly, its form factor vanishing on the reciprocal lattice, and the
 952 spectral construction of Eq. (35) is the periodic coupled-dipole method of
 953 Chaumet et al. [14] with the Galerkin double average in place of the point
 954 dipole. The second-order convergence reported for the DDA by Yurkin et al.
 955 [1] corresponds to the uniform-field voxel's, and the opposite-signed constants
 956 of Eq. (30) explain why a polarisability correction can improve one observable
 957 only at the expense of another, as the package's own form-factor corrections
 958 do in reflection and transmission. Going beyond second order requires a richer
 959 internal field, which is the point of higher-order bases for volume integral
 960 equations; the first-moment voxel is the lowest such basis, and in its spectral
 961 form the Galerkin double average makes the lattice sum converge without
 962 Ewald splitting.

963 *The tension between the two error sources.* The coupled-dipole method has
 964 two sources of error, the single site and the coupling between sites, and each
 965 has been corrected on its own: the polarisability by the Clausius–Mossotti
 966 relation, the lattice dispersion relation or the exact static limit, and the
 967 coupling by integrating or filtering the Green tensor. The two are not in-
 968 dependent. The point kernel's evanescent orders carry a static local term
 969 whose size per voxel volume does not shrink with the voxels (§4), and the
 970 single site must cancel exactly that term. A polarisability tuned for one
 971 kernel is therefore wrong for another: the lattice dispersion relation corrects
 972 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 973 tered one changes the term it must cancel. Tuning either alone moves the
 974 error between the two, and refinement cannot reveal a mismatch, because
 975 the surviving term is scale-invariant. For space-filling cubes the tension is

976 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 977 term that the closure carries (Eq. (29)), so the coupling error vanishes iden-
 978 tically and every remaining error is the single site's, the variation of the field
 979 across the cell. That error has a second tension of its own. It is a midpoint-
 980 rule error whose constants have opposite signs in reflection and transmission
 981 (Eq. (30)), so any scalar adjustment of the single site that improves one wors-
 982 ens the other; the package's far-field corrections reduce the reflection error
 983 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 984 however chosen, can remove it. What removes it is a richer internal field:
 985 the first moment lowers the error by two orders in kd in both observables at
 986 once.

987 *The cost of fourth order.* A first-moment voxel carries four times as many
 988 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 989 forms show what that buys. To reach an error ε in reflection, the uniform
 990 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 991 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 992 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 993 number of unknowns by about 5×10^4 (these are the Born-order constants;
 994 at the contrast studied the first-moment voxel's measured errors lie within
 995 5% of them, while the uniform voxel's transmission constant grows with the
 996 contrast, so the comparison is if anything conservative). The stratified model
 997 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 998 moment voxels for the same accuracy. Higher moments continue the trend
 999 in one dimension: by Eq. (34) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 1000 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already

1001 a quarter of a wavelength. In three dimensions, with the Legendre products
1002 of total degree at most p , a voxel carries 10 ($p = 2$) or 20 ($p = 3$) weights per
1003 state component, against 4 for the first-moment voxel. The voxel of degree
1004 2 converges at sixth order in the layer at oblique incidence (Table 6); that of
1005 degree 3 has not been built in three dimensions. The gain is largest precisely
1006 where it matters: when the model varies on scales that are coarse compared
1007 with the accuracy the wavefield must have.

1008 *Limitations and outlook.* The results concern laterally uniform media, a ho-
1009 mogeneous layer and a randomly stratified one, for which the discrete system
1010 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
1011 moment voxel in a full three-dimensional solver, where the gradient channels
1012 couple through the evanescent lattice orders (§8) and the tiling identity holds
1013 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
1014 voxel converging at second order on a laterally heterogeneous body against
1015 an exact solution, and the companion paper [2] shows the first-moment voxel
1016 converging at fourth order there. A contrast that varies within a cell, which
1017 §9 shows is needed beyond second order in a smoothly varying medium, has
1018 a further consequence in three dimensions: its linear part is odd about the
1019 cell’s centre, so it couples the single site’s uniform-field and first-moment
1020 blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded
1021 voxel therefore needs at least the first-moment block; that voxel is built,
1022 and tested on the graded sphere, in the companion paper. The numerical
1023 evidence is at moderate, physically typical contrasts (up to 15% in density
1024 and moduli) and at $k_S h \leq 0.3$, the range in which the single-site T -matrix is
1025 validated; the closed forms are first order in the contrast, and the departures

from them at the full contrast are of second order. What is established is the order of convergence and its leading constant, not the behaviour at strong contrast, where the single site's own closure may limit the accuracy before the lattice coupling does. The single site's dynamic error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§7–8). Two extensions follow directly. The first is the three-dimensional solver itself. The second is a stratified *background*, a layered reference medium, as distinct from the stratified contrast of §9. There the lattice coupling then carries the reverberation of the layering, which is not translation invariant in depth, and whether a tiling identity holds for it is the natural next question.

12. Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree p and the contrast of the medium to degree r in each voxel, the scheme converges at order $\min(2p + 2, 2r + 2)$. In a uniform layer the leading error is $c_p(kd)^{2p+2}$ with the closed-form constant of Eq. (34), and a single plane of such voxels reproduces the thin-layer series term by term through $O(D^{2p+2})$; the orders 2, 4 and 6 of the degrees 0, 1 and 2 hold for P, SV and SH waves at normal and oblique incidence. The first-moment voxel's leading error, $-(kd)^4/720$, does not

1050 depend on the direction of observation, whereas the collocation scheme's is
 1051 opposite in sign in reflection and transmission, so that no scalar correction of
 1052 the single site removes it. Where the medium varies within a cell, a contrast
 1053 held constant caps every voxel at second order, the coupling for a contrast
 1054 of degree r needs nothing beyond the same kernel moments to degree $p + r$,
 1055 and the error of the term of second order in the contrast is the relative
 1056 projection error of the medium at degree $\min(p, r)$, exactly in the long-wave
 1057 limit. That error is the single site's: its T -matrix sums the rescattering
 1058 within a voxel exactly for the voxel as represented, and the represented voxel
 1059 is not the medium. The two bases must therefore be matched. Where the
 1060 medium is piecewise constant on the cells the projection error vanishes: the
 1061 orders survive the interfaces of a randomly stratified model, provided voxel
 1062 boundaries coincide with them, and there the first-moment voxel reaches an
 1063 accuracy of 10^{-6} with one voxel per model layer, where the uniform-field
 1064 voxel needs about forty. A sphere whose contrast falls smoothly to zero at
 1065 its surface converges at second order against its exact solution, as the layer
 1066 predicts, to 0.018% at $k_S a = 0.5$ with twenty-four voxels across; a sharp
 1067 sphere's error is set by its staircase, which changes with the grid.

1068 Three properties of the cube make the experiment exact. Because cubes
 1069 tile the plane, the source-cell-averaged coupling between voxels is the con-
 1070 tinuum's cell integral exactly, and at normal incidence the lattice kernel is
 1071 available in closed form with no lattice sum. The single site is derived rather
 1072 than assumed: the moments of the Green's tensor over the cube, taken as
 1073 distributions, give a closed hierarchy whose first-gradient truncation is the
 1074 uniform-strain closure. That closure is exact for a sphere in the static limit,

1075 it departs from exact scattering by a sphere only by a real $(ka)^2$ term, with
1076 the radiation damping exact, and for a cube it differs from the sphere only
1077 by a cubic anisotropy of the shear response. And the collocation closure
1078 carries the same self term as the lattice kernel, which cancels, so the discrete
1079 system is a collocation of the continuum equation, whose second-order error
1080 is a midpoint-rule error of the internal field, known in closed form for ev-
1081 ery discretisation. The results, and the evidence for each, are collected in
1082 Table 11.

1083 *The error is known in advance.* For every discretisation studied here the
1084 error is not something to be discovered by refining until the answer stops
1085 changing. Its order and its leading constant are known before anything is
1086 computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the
1087 collocation voxel (Eq. (30)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the
1088 voxels of degree 0, 1 and 2 (Eq. (34)), and the order $\min(2p + 2, 2r + 2)$
1089 where the medium varies within a cell (Eq. (37)), with the size of the error
1090 there given by the relative projection error of the medium (Eq. (39)), which
1091 is computed from the model and the grid alone. That is worth more than
1092 the order alone, for three reasons. First, the cell size and the degree of the
1093 voxel that a required accuracy needs follow from the wavenumber and the
1094 medium, before the computation and without a refinement study. Secondly,
1095 it turns a refinement study from a hope into a test. A scheme of this kind can
1096 converge smoothly to the wrong limit (§4), and smooth convergence alone
1097 cannot reveal it; but the difference between two grids must have the known
1098 order and the known constant, and a computation whose differences do not
1099 is in error from some cause other than the discretisation. Thirdly, it shows

Table 11: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (21)	sum rule (22) exact; M, Q equal an independent archive, 81/81
Channels (23)–(25)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (26)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (27)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (29)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (30)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (32)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (35)	order 4, P and S, 20° and 40°

1100 where effort pays. The error is that of representing the field and the medium
1101 within a cell, so raising the degree of the cell does what multiplying the cells
1102 cannot: in the stratified model one first-moment voxel per layer reaches 10^{-6}
1103 where the uniform-field voxel needs about forty. Whether the error of a body
1104 with three-dimensional structure can be known in advance in the same way
1105 is the question taken up in the companion paper [2].

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1107 This work began thirty years ago in the author's doctoral research at
1108 the Research School of Earth Sciences of the Australian National University,
1109 Canberra [17].

1110 **Declaration of competing interest**

1111 The author declares no known competing financial interests or personal
1112 relationships that could have appeared to influence the work reported in this
1113 paper.

1114 **Declaration of generative AI and AI-assisted technologies in the** 1115 **manuscript preparation process**

1116 During the preparation of this work, the author used Claude (Anthropic)
1117 in order to check the existing Mathematica notebooks, Python and Fortran
1118 code, to build tests to ensure their correctness, and to write the TikZ code
1119 of the diagrams. After using this tool, the author reviewed and edited the
1120 content as needed and takes full responsibility for the content of the published
1121 article.

1122 **Data availability**

1123 The Mathematica notebooks and Python scripts that derive and check
1124 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)
1125 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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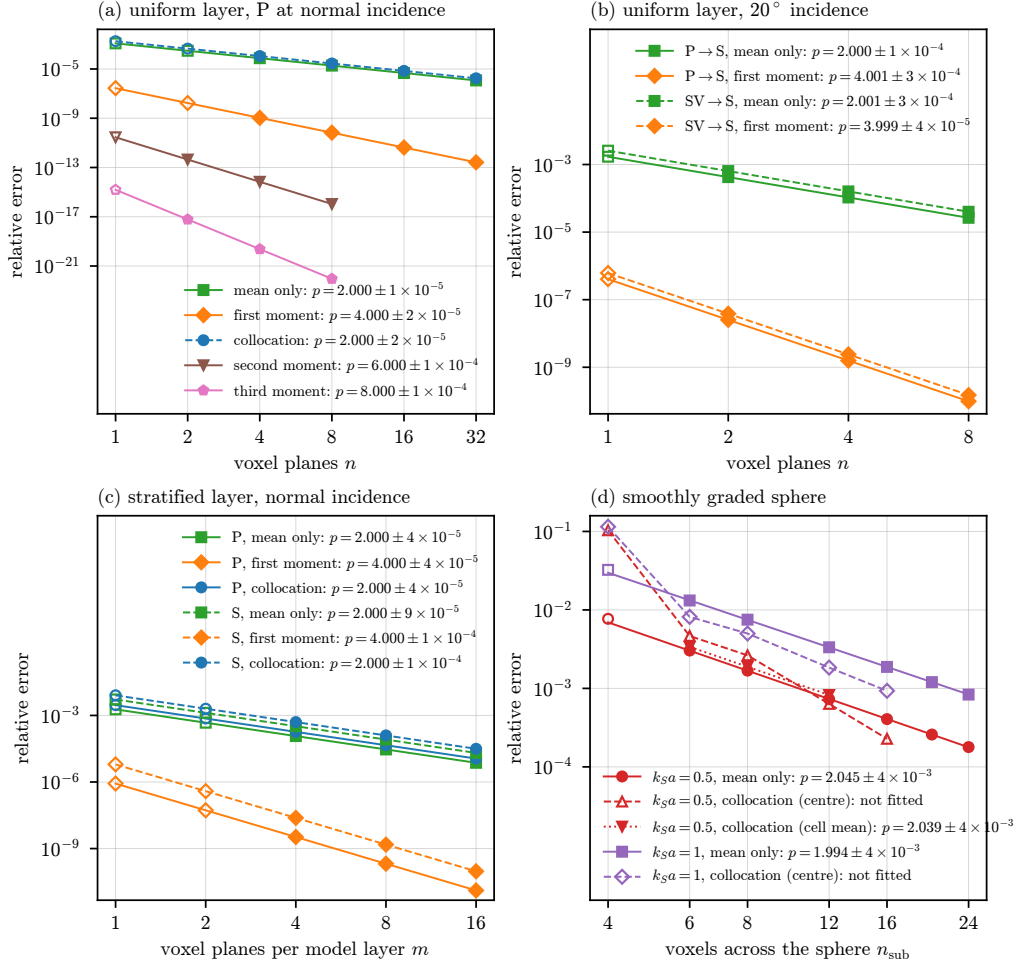


Figure 5: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree $p = 1$ (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (32) and (35)). (a) The uniform layer, P reflection at normal incidence ($D = 2\text{ m}$, $\omega = 300\text{ rad s}^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor $3/2$ of Table 3; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and